Course Syllabus Part 1: Course Specific Information

ELEC-8330 (Computational Intelligence)

Faculty of Engineering, Department of Electrical and Computer Engineering

University of Windsor, Canada

Semester: Winter 2024

Instructor information

- Name: Hon Kwan, PhD, FIET, PEng
- Office: Online
- Office Hours: 11:20 AM to 11:50 AM, Monday and Wednesday, face-to-face immediate after class. Online Microsoft Team meeting during 2:00 PM to 3:00 PM Wednesday by email appointment 24 hours in advance for one or more 15-minute timeslots stating the question such as the topic and its page and line numbers on any of the 2 eBooks.
- Office Phone Number: 519-253-3000 x2569
- Email: kwan1@uwindsor.ca (24hr response time Mon.-Fri.)
- Mailbox: Department of Electrical and Computer Engineering
- Website: https://www.uwindsor.ca/isplab

Graduate Assistant (GA) information

<table>
<tr><th>Name</th><th>Office</th><th>Office Hours (and by appointment)</th><th>Office Phone Number (extension #)</th><th>Email (24hr response time Mon.-Fri.)</th><th>Mailbox</th></tr>
<tr><td rowspan="2">Yuheng Song</td><td>-</td><td>2 hours per week</td><td>-</td><td>song56@uwindsor.ca</td><td>ECE</td></tr>
<tr><td colspan="5">Responsible for online tutorial and office hours, and answering students’ emails on questions from Chapters 5, 8, 9, 10, 11 on the two eBooks.</td></tr>
<tr><td rowspan="2">Jiasong Liu</td><td>-</td><td>2 hours per week</td><td>-</td><td>liu2l1@uwindsor.ca</td><td>ECE</td></tr>
<tr><td colspan="5">Responsible for online tutorial and office hours, and answering students’ emails on questions from Chapters 6, 7, 13, 19, 20 on the two eBooks.</td></tr>
<tr><td colspan="6">Microsoft Team: GAs’ online tutorial and office hours are hosted via Microsoft Team meetings, ELEC8330-1-R-2024W. Students can login from 5:30 PM to 7:20 PM, Tuesday to seek GAs’ helps on course contents and assignments.
Emails: Immediate GAs’ helps on course contents and assignments are supported via emails which will be answered within 24 hours Monday to Friday.</td></tr>
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Class and lab information

- Class Location: Toldo Room 200
- Class Time: Monday & Wednesday 10:00 AM to 11:20 AM.
- Tutorial Location: Microsoft Teams, ELEC8330-1-R-2023W
- Tutorial Time: Tuesday, 5:30 PM to 7:20 PM
- Estimated division of Learning hours:
  - lecture: 3 hours/week
  - individual work: Minimum 5 hours/week (Study and solve examples and worked problems in eBooks. Study eBooks chapters and complete assignments and quizzes.)
  - class discussion: Attend tutorials, 2 hours/week

- Lecture: 3 hours/week
- Tutorial: 2 hours/week
- Credit weight: 4
- Course format: Face-to-face lecture
- Pre-requisites, from the current University of Windsor Undergraduate Calendar or Graduate Calendar (http://web4.uwindsor.ca/calendar):
    - Undergraduate elementary courses in linear algebra and calculus are assumed.

Course Description

From the current University of Windsor Graduate Calendar (http://web4.uwindsor.ca/calendar): Models of the human brain and sensory systems. Neural networks and learning algorithms. Fuzzy sets, fuzzy logic, and fuzzy systems. Evolutionary computation. Advanced topics in computational intelligence.

Resources

- Course Brightspace site: https://brightspace.uwindsor.ca/d2l/login
- Primary text (available for purchase from Publisher: https://www.dfisp.org)
    - H. K. Kwan, Intelligent Computing: A Computing Approach to Artificial Intelligence,
    - Edition 1.5, dfisp.org, 721 pages, 4 February 2020.
    - H. K. Kwan, MATLAB and Worked Problems in Intelligent Computing and System
    - Design, Edition 1.4, dfisp.org, 253 pages, 24 January 2019.
- Additional resources
    - N.A.
- Web resources
    - Keywords: N.A.
    - Organizations: N.A.

Evaluation Methods

The course grade will be evaluated as follows:

Method of Evaluation	**% of Final Grade** No assignment > 50% Senate Bylaw 54 - Paragraph 2.5.1	**Due Dates*** (Include how students will submit the assessment)	**Learning Outcomes covered in this assessment**
Assignments (Maximum 10 individual assignments)	20%	One week after the end of Chapter in Brightspace	1a, 1b, 1c, 2b, 7a
Midterm exam (Open eBook at Engineering Computing Lab, CE2105 & CE2191 or Toldo 200)	28% (Chapters 5,6,7,8)	Feb 28, Midterm Exam (9:55 AM to 11:25 AM)	1a, 1b, 1c, 2b, 7a
Final exam * (Open eBook at Engineering Computing Lab, CE2105 & CE2191 or a specified classroom)	42% (Chapters 9,10,11,13,19,20)	To be announced by Registrar	1a, 1b, 1c, 2b, 7a
Participation	10%	Held during	1a, 1b, 1c, 2b, 7a

(Maximum 1 quiz per week starting from 2nd week.)		Mon or Wed lecture class.	

***** Two to three-hour examination slots will normally be scheduled in the formal final examination periods in each semester for all courses which terminate in that semester. All final examinations shall take place (or fall due, as the case may be) during the two to three-hour final examination slot so scheduled. The actual duration of testing procedures during the scheduled final examination slot may be less than the scheduled time, at the discretion of the individual instructor. **Senate Bylaw 54 – Section 1.2**

Course Schedule

The following course schedule is approximate.

Week	**Date**	**Subject, activity, assignment, etc.**	**Textbook Chapter or Readings**
	Jan 8	Introduction (Syllabus, course overview)	Syllabus
	Jan 10	Associative memories	5
2	Jan 15	Associative memories	5
	Jan 17	Self-organization map	6
3	Jan 22	Self-organization map	6
	Jan 24	Self-organization map	6
4	Jan 29	Learning vector quantization	7
	Jan 31	Radial basis function networks	8
5	Feb 5	Radial basis function networks	8
	Feb 7	Fuzzy sets, Membership functions, Linguistic variables	9
6	Feb 12	Fuzzy sets, Membership functions, Linguistic variables	9
	Feb 14	Fuzzy sets, Membership functions, Linguistic variables	9
Reading Week – February 17-25, 2024			
7	Feb 26	Fuzzy rules and fuzzy reasoning	10
	Feb 28	Midterm Exam (9:55 AM to 11:25 AM)	5, 6, 7, 8
8	Mar 4	Fuzzy rules and fuzzy reasoning	10
	Mar 6	Fuzzy systems	11
9	Mar 11	Fuzzy systems	11
	Mar 13	Genetic Algorithms	13
10	Mar 18	Genetic Algorithms	13
	Mar 20	Convolutional Neural Networks	19

11	Mar 25	Convolutional Neural Networks	19
	Mar 27	Convolutional Neural Networks	19
12	Apr 1	Residual Learning and Networks	20
	Apr 3	Residual Learning and Networks	20
13	Apr 8	Residual Learning and Networks	20
	Apr 11	Final Exam period for all courses start	
		Final Exam	9,10,11,13,19,20

Learning Outcomes

In this course, students will…

Number	Learning Outcome	Learning Outcome Code (i.e., 1a)*
1	Recognize the terminologies and describe the knowledge in computational intelligence and apply mathematical skills to solve questions in assignments, midterm examination and final examination.	1a
2	Understand the principles and techniques of computational intelligence, apply and integrate knowledge to solve questions in assignments, midterm examination and final examination.	1b
3	Differentiate the three major domains of computational intelligence, namely, neural, fuzzy, and evolutionary computing; and their core topics.	1c
4	Carry out analysis and evaluation from the information of a question in computational intelligence to reach a solution in assignments, midterm examination and final examination.	2b
5	Access, retrieve, and evaluate information, and define the problems and work-out the solutions of open-ended problems in computational intelligence.	3b
6	Design simple systems using neural, fuzzy, and evolutionary techniques through examples and problems for applications.	4c
7	Use MATLAB programming tools to simulate solutions for problems in computational intelligence.	5c
8	Present in an organized manner the solutions of questions in assignments, midterm examination and final examination.	7a

* Learning Outcome Codes are keyed to the Table of Graduate Attributes and Indicators, which appears in Part 2 of the course syllabus

CEAB Hours

Subject Areas	**Accreditation Units** One hour of lecture (corresponding to 50 minutes of activity) = 1AU One hour of laboratory or scheduled tutorial = 0.5 AU
Mathematics	1 AU/week (25%)
Natural Sciences	-
Engineering Science	2 AU/week (50%)

Engineering Design	1 AU/week (25%)
Complementary Studies	-

Other electronic devices aside from calculators

☒ Electronic devices aside from calculators are **NOT** permitted during tests/exams.

☐ Other electronic devices aside from calculators are permitted during tests/exams. Acceptable electronic devices include: _Nil___________

Calculators

- Approved calculator: Calculators approved by the Faculty of Engineering are Casio: fx-991W; Sharp: EL531M, EL531Q, EL509VHB or EL531VB; Texas Instruments: TI-32, TI-32solar, TI-30X, TI-30XA (solar or battery), TI30X-IIS or TI30X-IIB, TI34II, TI36X solar. Additional calculators approved by the Instructor are Casio: fx-82MS, fx-100MS, fx-115MS, fx-300MS.

Laboratory Experience

Will there be a laboratory experience and safety procedures instruction? ☐ Yes ☒No

Student Perceptions of Teaching survey

The Student Perceptions of Teaching survey will be administered during the last two weeks of classes, for 12-week courses. The Student Perceptions of Teaching survey will be administered during the last week of classes, for 6-week courses.

Use of Plagiarism-Detection Software in This Course

☐ Plagiarism-detection software, [*insert specific software name*], will **NOT** be used in this course.

☒ Plagiarism-detection software, [*insert specific software name*], may be used in this course.

Course Syllabus Part 2: Faculty of Engineering Information

The Faculty's Commitment to Reconciliation, Equity, Diversity, and Inclusion

The Faculty of Engineering follows the lead of Canada's Engineering Profession with its commitment to equity, diversity, inclusivity, and reconciliation as addressed in language from the Profession's 2009 Montreal Declaration.

While the profession of engineering itself is largely invisible, its impact is visible all around us: in the built environments of our cities and towns; in our infrastructure; in our technology; in the ways we work and the systems we rely on to remain safe and secure. As a profession, we are committed to helping provide the best possible quality of life for all Canadians, with the understanding that it is the international measure of Canada.

We, Canada's engineers,

- *Pledge to make educational enhancements that will encourage broader participation in the profession by all segments of the population and foster innovation.*
- *Acknowledge that we must encourage the greater participation of underrepresented groups such as Aboriginal Peoples.*
- *Acknowledge that we must attract and retain women in much greater numbers.*
- *Need to be more socially aware to address the unique issues facing individuals in our society.*
- *Understand that collaboration with First Nations, Metis, and Inuit people will be essential to seizing development and economic opportunities across Canada.*

Further the Faculty of Engineering acknowledges its commitment to the outcomes of Canada's Truth and Reconciliation Commission. It continues its efforts to include *"curriculum on residential schools, Treaties, and Aboriginal peoples' historical and contemporary contributions to Canada"* in the program of every student.

The Faculty of Engineering promotes the recognition that "the University of Windsor sits on the traditional territory of the Three Fires Confederacy of First Nations, comprised of the Ojibwa, the Odawa, and the Potawatomi. We respect the longstanding relationships with First Nations people in this place in the 100-mile Windsor-Essex peninsula and the straits – les détroits – of Detroit."

The Faculty of Engineering supports efforts by its students, staff, and faculty members in their recognition of September 30 as the National Day for Truth and Reconciliation, and December 6 as the National Day of Remembrance and Action on Violence Against Women.

Information for Students about Course Procedures

Assessment Considerations

- **Submission of Assignments**
    - All assignments will be submitted electronically through the course Brightspace site.

- **Late assignments, reports, or projects**
    - It is expected that students who are experiencing difficulty meeting a deadline will contact the course instructor as soon as possible to discuss the situation in advance of the deadline.
    - No late assignments will be accepted, and the student will receive grade 0.

- **Missed Quizzes, Assignments, Midterm Exam, or Final Exam**
    - All missed Quizzes, Assignments, Midterm Exam, or Final Exam will receive grade 0.

- **Late Registration into Course**
    - Students who register late for the course are responsible to familiarize themselves with course information that they missed prior to registration. No special accommodation will be provided for missed assignments/assessments.

Important Dates

References are made to Senate Bylaw 54, which can be found at *lawlibrary.uwindsor.ca/Presto/home/home.aspx*

January 8, 2024	**First day of classes** - The instructor must provide students with a course outline (hard-copy or electronic) as per **Senate Bylaw 54 – Paragraph 2.1.**
January 21, 2024	The last date to **ADD/DROP** a course or change sections is two weeks after the start of classes. Last day for changes to the course syllabus per **Senate Bylaw 54 – Paragraph 2.7**. Compelling reasons can allow for changes after this date; students must receive 2 weeks notice.
January 28, 2024	For Winter 2024 courses, the last day for students to make a formal request to instructor(s) for **accommodation for missed mandatory academic events** (tests, midterms, labs) **due to Religious Observance or attendance at a recognized University-sponsored event, or** should be done within the first three weeks of the academic term.
January 28, 2024	For Winter 2024 courses, the last day for students to make a formal request to instructor(s) for accommodation for **three or more major in-term evaluations** scheduled or due within a 24-hour period. **Senate Bylaw 54 – Paragraph 2.5.3**
February 5, 2024	**Financial Drop Date** – Last day to receive full-tuition refund for Winter 2024 courses (less non-refundable deposit if applicable). Any Winter 2024 course dropped after this date will receive 0% refund.
February 17-25, 2024	**Reading Week** for Winter 2024 courses – No forms of assessment shall be scheduled or due. **Senate Bylaw 54 – Paragraph 2.3**

February 19, 2024	Family Day – University is closed. No forms of assessment shall be scheduled or due. **Senate Bylaw 54 – Paragraph 2.3**
February 23, 2024	University offices are closed. No forms of assessment shall be scheduled or due. **Senate Bylaw 54 – Paragraph 2.3**
February 28, 2024	**Application Deadline for Alternative Final Examination(s) Due to Conflict with Religious Conviction** for Winter 2024 courses. **Senate Bylaw 54 – Paragraph 2.22**
February 28, 2024	**Application Deadline for Alternative Final Examination(s)** Due to 3 Exams Scheduled on the Same Day or over a 24-hour period for Winter 2024 courses. **Senate Bylaw 54 – Paragraph 2.5.2**
March 15, 2024	Deadline for instructors to provide meaningful feedback on student performance, constituting a minimum of 20% of the final grade, unless exempted by the Dean with the instructor's statement of rationale included as part of this course syllabus. **Senate Bylaw 54 – Paragraph 2.6**
March 17, 2024	Last day to **voluntarily withdraw** from Winter 2024 courses. After this date, students remain registered in the course and receive a final grade as appropriate.
March 29, 2024	Good Friday – University is closed. No forms of assessment shall be scheduled or due. **Senate Bylaw 54 – Paragraph 2.3**
April 2-8, 2024	**The last 7 calendar days** prior to, and including, the last day of classes must be free from any procedures for which a mark will be assigned, including the submission of assignments such as essays, term papers, and take-home examinations per **Senate Bylaw 54 – Paragraph 1.3** **Engineering courses that have a regularly scheduled laboratory or tutorial are exempted by the Dean** when the tutorial or laboratory assignment is begun, completed, and submitted within the regularly scheduled class time.
April 8, 2024	Make up date for Good Friday holiday (March 29)
April 8, 2024	**Last day of classes** for Winter 2024 courses.
April 9-10, 2024	**Reading period** prior to final exams. No forms of assessment shall be scheduled or due.
April 11-22, 2024	**Final examination period** for Winter 2024 courses.
April 23, 2024	**Alternate Final Exams Day** for Winter 2024 courses.
May 6, 2024	**First day of Classes** for Summer/Intercession 2024 courses.

As per **Senate Bylaw 54 – Paragraph 2.11**, a student who believes that a provision of paragraphs 2.1 – 2.10 is being violated is encouraged to resolve the matter informally with the instructor and/or the AU Head. If the complaint is not resolved, the student may appeal to the Dean of the Faculty.

Canadian Engineering Accreditation Board (CEAB) Graduate Attributes (1 - 12)
University of Windsor - Faculty of Engineering Indicators (a, b, c)

CEAB Graduate Attributes and Indicators
1. A knowledge base for engineering *Demonstrated competence in University level mathematics, natural sciences, engineering fundamentals, and specialized engineering knowledge appropriate to the program.* a) Demonstrate competence in mathematics and modeling. b) Demonstrate competence in natural sciences and engineering fundamentals. c) Demonstrate competence in specialized engineering knowledge appropriate to the program.
2. Problem analysis *An ability to use appropriate knowledge and skills to identify, formulate, analyze, and solve complex engineering problems in order to reach substantiated conclusions.* a) Classify a given problem according to commonly used solution methods. b) Recognize given and missing information, assumptions, and information to be gathered for the solution method. c) Execute a problem solution and interpret the results.
3. Investigation *An ability to conduct investigations of complex problems by methods that include appropriate experiments, analysis and interpretation of data, and synthesis of information in order to reach valid conclusions.* a) Explain why an experimental methodology is appropriate for a given problem. b) Conduct an experiment. c) Interpret experimental results to formulate valid conclusions.
4. Design *An ability to design solutions for complex, open-ended engineering problems and to design systems, components or processes that meet specified needs with appropriate attention to health and safety risks, applicable standards, economic, environmental, cultural and societal considerations.* a) Generate a problem statement and its design objectives. b) Consider constraints/stakeholders (e.g., health and safety, codes and standards, economics, and environmental, social, and cultural considerations) when selecting a final design from a diverse set of candidate solutions. c) Refine and advance a design to its final end state.
5. Use of engineering tools *An ability to create, select, apply, adapt, and extend appropriate techniques, resources, and modern engineering tools to a range of engineering activities, from simple to complex, with an understanding of the associated limitations.* a) Select, create, modify, use, and understand the limitations of computational and analytical methods to model and analyze engineering systems. b) Select, create, modify, use, and understand the limitations of measuring instruments and testing equipment to collect data for analysis.
6. Individual and teamwork *An ability to work effectively as a member and leader in teams, preferably in a multi-disciplinary setting.* a) Define individual contributions to the team effort. b) Employ interpersonal skills to promote team dynamics. c) Integrate individual contributions into a coherent team report or presentation.
7. Communication skills *An ability to communicate complex engineering concepts within the profession and with society at large. Such ability includes reading, writing, speaking and listening, and the ability to comprehend and write effective reports and design documentation, and to give and effectively respond to clear instructions.* a) Comprehend and compose engineering-based written communications both from and for a variety of audiences. b) Comprehend and deliver engineering-based oral communications both from and for a variety of audiences. c) Prepare, integrate and interpret graphical communications used in written and visual formats (Examples: data depicted through graphs, charts, and tables; other engineering drawings).
8. Professionalism *An understanding of the roles and responsibilities of the professional engineer in society, especially the primary role of protection of the public and the public interest.* a) Describe the role of the engineer in protecting and promoting the public welfare both locally and globally. ***b)*** Demonstrate professional behavior in their individual interactions with others (Examples: proper etiquette in e-mail and other communications, adherence to submission deadlines, courteous interactions with students and staff).

9. Impact of engineering on society and the environment *An ability to analyze societal and environmental aspects of engineering activities. Such ability includes an understanding of the interactions that engineering has with the economic, health, safety, legal, and cultural aspects of society, the uncertainties in the prediction of such interactions; and the concepts of sustainable design and development and environmental stewardship.* a) Demonstrate an awareness of legal issues relevant to engineering activity. b) Identify the impacts of engineering activity on society and the environment. c) Identify ways to mitigate the potential negative impact of engineering activities on society and the environment.
10. Ethics and equity *An ability to apply professional ethics, accountability, and equity.* a) Define the concepts of ethics and equity. b) Apply aspects of the PEO Code of Ethics to their current studies. c) Identify equity issues within both the engineering profession and Canadian society, with an emphasis on the role of Aboriginal peoples, women, visible minorities, persons with disabilities, and sexual minorities.
11. Economics and project management *An ability to appropriately incorporate economics and business practices including project, risk and change management into the practice of engineering and to understand their limitations.* a) Evaluate the economic and financial performance of an engineering activity, including life-cycle costs and benefits. b) Estimate, organize, and manage engineering activities to be within time and budget constraints.
12. Life-long learning *An ability to identify and to address their own educational needs in a changing world in ways sufficient to maintain their competence and allow them to contribute to the advancement of knowledge.* a) Identify the benefits of becoming a member of a professional society. b) Independently summarize, analyze, synthesize, and evaluate information from a wide variety of sources, including library methods, relevant codes/standards/regulations, and digital methods.

Grading

Grades for the course will be consistent with the following table, per the University of Windsor Policy on Grading and Calculation of Averages.

☐ Undergraduate Course:

Letter	A+	A	A-	B+	B	B-	C+	C	C-	F
% Range	90-100	85-89.9	80-84.9	77-79.9	73-76.9	70-72.9	67-69.9	63-66.9	60-62.9	0-59.9

Feeling Overwhelmed?

From time to time, students face obstacles that can affect academic performance. If you experience difficulties and need help, it is important to reach out to someone.

For help addressing mental or physical health concerns on campus, contact (519) 253-3000:
- Student Health Services at ext. 7002 (http://www.uwindsor.ca/studenthealthservices/)
- Student Counselling Centre at ext. 4616 (http://www.uwindsor.ca/studentcounselling/)
- Peer Support Centre at ext. 4551

24 Hour Support is Available

- My Student Support Program (MySSP) is an immediate and fully confidential 24/7 mental health support that can be accessed for free through chat, online, and telephone. This service is available to all University of Windsor students and offered in over 30 languages. Call: 1-844-451-9700, visit https://keepmesafe.myissp.com/ or download the My SSP app: Apple App Store/Google Play.

A full list of on- and off-campus resources is available at http://www.uwindsor.ca/wellness.

Should you need to request alternative accommodation contact your instructor or associate dean.

Services Available to Students at the University of Windsor

Students are encouraged to discuss any disabilities, including questions and concerns regarding disabilities, with the course instructor. Let's plan a comfortable and productive learning experience for everyone. The following services are also available to students:

- Sexual Misconduct Response & Prevention Office: http://www.uwindsor.ca/sexual-assault
- Student Accessibility Services: http://www.uwindsor.ca/studentaccessibility/
- Skills to Enhance Personal Success (S.T.E.P.S): http://www.uwindsor.ca/lifeline/steps-skills-to-enhance-personal-success
- Student Counseling Centre: http://www.uwindsor.ca/scc
- Academic Advising Centre: http://www.uwindsor.ca/advising/
- Writing Support Desk: https://www.uwindsor.ca/success/318/writing-support-desk
- Information Technology Services: https://www.uwindsor.ca/itservices/support
- Student Health Services: https://www.uwindsor.ca/studenthealthservices/
- Mental Health: https://www.uwindsor.ca/wellness

Student Self Report of Illness

Medical or Compassionate Absences: If students will miss an exam, class, test, assignment etc. and are requesting an accommodation, they must report the illness to engadmin@uwindsor.ca, along with the appropriate documentation (e.g., a doctor's note for an illness) within 72 hours of the missed assessment. Determinations about whether and how to accommodate students who submit requests for consideration based on compassionate grounds will, as usual, be made by instructors and/or the Associate Dean, in keeping with any standard procedures within specific Faculties and the Senate bylaws.

Students should report a COVID related illness or isolation/quarantine to COVID19reporting@uwindsor.ca. The email will generate an automated response with instructions.

Minimum technology requirements

To support your studies, you will require access to particular computer hardware and software for most UWindsor courses. The UWindsor standard computing platform supported by IT Services is a device running current, supported versions of Microsoft Windows and MS Office 365. For detailed recommendations, please read this FAQ: http://ask.uwindsor.ca/app/answers/detail/a_id/688

General Class Expectations

Attendance and punctuality

- Attendance in classes and labs is critical to student success; students should seize the opportunity to share and discuss information in labs, tutorials, and classes. The course is designed to move swiftly and efficiently. If a student is going to miss a class or lab, s/he should inform the instructor and GA before missing the class or lab.

Communication

- Students are encouraged to utilize office hours to ask questions. **Only emails sent from a uwindsor email address will be responded to.** Emails should be sent with courtesy; they should include an informative subject line, a salutation (e.g., Hello Dr. Name), a body, and a closing (e.g., Best regards, Name).

Group work

- Groups are encouraged to develop ground rules, identify roles and responsibilities, set timelines, and set standards of communication for the group.

Academic Integrity

All incidents of academic dishonesty will be documented with the Associate Dean of Engineering – Academic. University procedures will be followed. Such incidents may include, but are not limited to: submission of assignments other than your own, receiving or sharing prior knowledge of test questions, sharing or receiving information during a test by any means (including electronic), possession of any electronic device (including cell phones) during a test except for an approved calculator, sharing or receiving knowledge of a test with students who have not yet written the test, sharing a calculator or formula sheet during the test, using a solutions manual to prepare submitted assignments.

Associated with on-line instruction and evaluation, the course instructor may identify academic integrity concerns with submissions for a graded aspect of the course. In such cases, the faculty member can set up an on-line meeting with individual student(s) to further assess knowledge in the given area. This on-line assessment can either confirm the original mark, or can be considered in place of the initial assessment to increase or decrease the original mark. All such cases will be documented with the Department Head.

The uploading of the required course eBooks, midterm exam, final exam, assignments, quizzes, laboratory, and project questions or prompts to, as well as the downloading of answers or responses from ChatGPT and other on-line services is a breach of academic integrity. Academic integrity violations will be dealt with according to Bylaw 31. Typical sanctions for a first offence range from a zero grade to a formal censure listed on your transcript.

Definition of Plagiarism

Source: *Student Code of Conduct*

Plagiarism: the act of copying, reproducing or paraphrasing portions of someone else's published or unpublished material (from any source, including the internet), without proper acknowledgement. Plagiarism applies to all intellectual endeavours: creation and presentation of music, drawings, designs, dance, photography and other artistic and technical works. In the case of oral presentations, the use of material that is not one's own, without proper acknowledgment or attribution, constitutes plagiarism and, hence, academic dishonesty. (Students have the responsibility to learn and use the conventions of documentation as accepted in their area of study.)

Use of Plagiarism-Detection Software

1. *Rationale.* The University believes in the right of all students to be part of a University community where academic integrity is expected, maintained, enforced, and safeguarded; it expects that all students will be evaluated and graded on their own individual work; it recognizes that students often have to use the ideas of others as expressed in written, published, or unpublished work in the preparation of essays, assignments, reports, theses, and publications. However, it expects that both the data and ideas obtained from any and all published or unpublished material will be properly acknowledged and sources disclosed. Failure to follow this practice constitutes plagiarism. The University, through the availability of plagiarism-detection software, desires to encourage responsible student behaviour, prevent plagiarism, improve student learning, and ensure greater accountability.

2. *Procedure*. [Plagiarism-detection software] may be used for some or all student assignments in this course, at the instructor's discretion. You may be asked to submit your assignments to the instructor in electronic form who will then submit the assignments to plagiarism-detection software if deemed necessary. (NOTE: this depends on the plagiarism checking tool)

3. *Privacy and Copyright.* Your privacy is protected even if your name and/or student number is on your assignments because the plagiarism-detection software does not make students' assignments available to outside third parties. Further, you retain the copyright in your work. Copyright, in relation to a work, is defined in Canada's Copyright Act, R.S.C. 1985, c. C-42, s. 3(1), which is available on the Department of Justice Canada website. Plagiarism-detection software use of student work complies with Canadian copyright and privacy laws.

4. *Originality Reports.* If the results of an originality report may be used to charge you with academic misconduct, you will be notified of the result of the report, and you will be given the opportunity to respond before any disciplinary penalty is imposed.

5. Plagiarism. Information about plagiarism and appropriate acknowledgement of sources can be found at the Office of Academic Integrity: http://www1.uwindsor.ca/academicintegrity/

Instructor's Policy on Recording Lectures

Lectures in the virtual classroom will not be recorded. Students are not permitted to record the lectures.

Students who record a lecture after the instructor has prohibited such recordings, or who record a guest lecturer or classmate presentation or performance without the written consent of the presenter, or who disseminate a recording without the explicit written permission from the instructor or presenter will be subject to the University's misconduct policies, at minimum.

Intellectual Property

Lectures and course materials prepared by the instructor are considered by the University to be an instructor's intellectual property covered by the Copyright Act, RSC 1985, c C-42. **The two eBooks (textbooks) are made available to you through individual purchase from the publisher (dfisp.org) for your own study purposes.** These materials cannot be shared outside of the class or "published" in any way. Lectures, whether in person or online, cannot be recorded without the instructor's permission. Posting course materials or any recordings you may make to other websites without the express written permission of the instructor may constitute copyright infringement.

Bylaws and Policies

The following are links to the University of Windsor bylaws and policies. The intention is to share these policies and bylaws with engineering students in a way that is straightforward and clear – because our learning depends on our ability to create an environment and culture that supports our individual and collective needs for learning and teaching.
University senate bylaws can be found: http://www.uwindsor.ca/secretariat/49/senate-bylaws
University senate policies can be found: http://www.uwindsor.ca/secretariat/48/senate-policies